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Introduction to the Course  
Fundamentals of Data Manipulation  
Week 1 - Assignment

- Quiz: Quiz 1 30 min
- Programming Assignment: Assignment 1 3h

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1. "What will be the output of the following code?" 1/1 point

```
1 import re
2 string = 'bat, lat, mat, bet, let, met, bit, lit, mit, bot, lot, mot'
3 result = re.findall('b[a-z]', string)
4 print(result)
```

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- ☐ ['bat','bet','bit','bot']
- ☒ ['bat','bot']
- ☐ 'bat, bot'
- ☐ 'bat, bet, bit, bot'

Correct [a-z] means any character from 'a' and 'z' (or both) hence only 'bat' and 'bot' would be extracted

2. 1/1 point

$$L_2 = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

Assume **a** and **b** are two (20, 20) numpy arrays. The L2-distance (defined above) between two equal dimension arrays can be calculated in python as follows:

```
1 def l2_dist(a, b):
2     result = ((a - b) * (a - b)).sum()
3     result = result ** 0.5
4     return result
```

Which of the following expressions using this function will produce a different result from the rest?

- ☐ l2\_dist(a, b)
- ☐ l2\_dist(np.reshape(a, (20 \* 20)), np.reshape(b, (20 \* 20)))
- ☒ l2\_dist(np.reshape(a, (20 \* 20)), np.reshape(b, (20 \* 20, 1)))
- ☐ l2\_dist(a.T, b.T)

Correct The ndim of the two inputs in D are different.

3. Consider the following variables in Python: 1/1 point

```
1 a1 = np.random.rand(4)
2 a2 = np.random.rand(4, 1)
3 a3 = np.array([[1, 2, 3, 4]])
4 a4 = np.arange(1, 4, 1)
5 a5 = np.linspace(1, 4, 4)
```

Which of the following statements regarding these variables is correct?

- ☒ a5.shape == a1.shape
- ☐ a1.shape == a2.shape
- ☐ a3.shape == a4.shape
- ☐ a4.ndim() == 1

Correct This is a correct expression because the two arrays have the same shape.

4. Which of the following is the correct output for the code given below? 1/1 point

```
1 import numpy as np
2
3 old = np.array([[1, 1, 1], [1, 1, 1]])
4 new = old
5 new[0, :2] = 0
6
7 print(old)
```

- ☐ [[0 1 1][0 1 1]]
- ☒ [[0 0 1][1 1 1]]
- ☐ [[1 1 0][1 1 0]]
- ☐ [[1 1 1][1 1 1]]

Correct Array slices are passed by reference; After the statement 'new=old', any changes made to 'new' will carry over to 'old', so 'old' will not remain unchanged

5. Given the 6x6 NumPy array **r** shown below, which of the following options would slice the shaded elements? 1/1 point

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 0  | 1  | 2  | 3  | 4  | 5  |
| 6  | 7  | 8  | 9  | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 |
| 18 | 19 | 20 | 21 | 22 | 23 |
| 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 | 32 | 33 | 34 | 35 |

- ☒ r[2:4,2:4]
- ☐ r[[2,3],[2,3]]
- ☐ r[[2,4],[2,4]]
- ☐ r[2:3,2:3]

Correct Array indices start with 0 and an array slice from m:n includes elements from indices m to n-1.

6. 1/1 point

```
1 import re
2 s = 'ACBCAC'
```

For the given string, which of the following regular expressions can be used to check if the string starts with 'AC'?

- ☐ re.findall('^AC', s)
- ☐ re.findall('[^A]C', s)
- ☒ re.findall('^AC', s)
- ☐ re.findall('AC', s)

Correct Here, the caret ^ denotes the beginning of the string, and hence will extract the first 'AC' from the string s.

7. What will be the output of the variable **L** after the following code is executed? 1/1 point

```
1 import re
2 s = 'ACAABAACAAB'
3 result = re.findall('A(1,2)', s)
4 L = len(result)
```

- ☒ 5
- ☐ 12
- ☐ 8
- ☐ 4

Correct The pattern we are using is a single A or a double A, hence we will find the results as (A)C(AA)B(AA)C(AA)(A)B.

8. Which of the following is the correct regular expression to extract all the phone numbers from the following chunk of text: 1/1 point

```
1 Office of Research Administration: (734) 647-6333 | 4325 North Quad
2 Office of Budget and Financial Administration: (734) 647-8844 | 309 Maynard, Suite 205
3 Health Informatics Program: (734) 763-2285 | 333 Maynard, Suite 500
4 Office of the Dean: (734) 647-3576 | 4322 North Quad
5 UMSI Engagement Center: (734) 763-1251 | 777 North University
6 Faculty Administrative Support Staff: (734) 764-9376 | 4322 North Quad
```